

Mohammad Ibrahim Khan

416-573-0020 | mohammadikhan821@gmail.com | [linkedin.com/in/mohammad-khan-888a691b8](https://www.linkedin.com/in/mohammad-khan-888a691b8) | github.com/mohammadikhan

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, C, C++

Frontend: React, Next.js, Angular, TailwindCSS, HTML/CSS

Backend: Node.js, FastAPI, Express.js, Spring Boot, .NET, Django, REST APIs

Cloud & DevOps: AWS, Docker, Kubernetes, Linux, Git, CI/CD (GitHub Actions)

Databases: PostgreSQL, MongoDB, MySQL, pgvector

AI/ML: RAG, LangGraph, Hugging Face Transformers, OpenAI API

Networking & Systems: TCP/IP, DNS, HTTP/HTTPS, Load Balancing, VPNs, Linux Administration, Network Troubleshooting, Client-Server Architecture

Certifications: AWS Certified Cloud Practitioner, AWS Certified Developer - Associate, Kubernetes and Cloud Native Associate (KCNA), HashiCorp Terraform Associate, LPI Linux Essentials

PROJECTS

VeriNews AI – AI Fake News Detector | *Python, FastAPI, React, Node.js, MongoDB* Jun. 2025 – Apr. 2026

- Achieved **92%** classification accuracy on **50,000+** news articles by fine-tuning Meta AI's **RoBERTa-base** model with **PyTorch** and **Hugging Face Transformers** to enable real-time misinformation detection
- Enabled scalable AI-powered article analysis by designing and deploying a three-service production architecture (**React, Node.js REST API, FastAPI ML inference**) across **Vercel** and **Render**
- Improved transparency and security by integrating **LIME** explainability, **JWT** authentication with refresh token rotation, and a **MongoDB** system for user feedback and model retraining

HoopIQ – Hybrid RAG NBA Knowledge Base | *Java, Spring Boot, PostgreSQL* May. 2026 – Jun. 2026

- Enabled accurate responses to both statistical and narrative NBA questions by combining **pgvector** semantic retrieval with **LLM-generated SQL** queries in a hybrid **RAG** pipeline
- Reduced retrieval latency to under **200ms** by optimizing **pgvector** indexes and document chunking strategies within **PostgreSQL**
- Decreased hallucinated statistics by implementing intelligent query routing between semantic search and structured database retrieval using **Spring AI**

TrailMind – Agentic Cycling Trip Planner | *Python, FastAPI, PostgreSQL, LangGraph* Apr. 2026 – Jun. 2026

- Automated personalized cycling itineraries, packing lists, and trail recommendations by orchestrating multiple **LLM agents** through a **LangGraph** workflow with **OpenAI API**
- Reduced hallucinated recommendations by grounding responses in **15+** Ontario trail documents and live web search using **pgvector** retrieval and external search tools
- Enabled serverless scalability by containerizing and deploying the backend on **AWS Lambda** and **API Gateway** using **Docker**

EXPERIENCE

Software Developer (Co-Op) Jan. 2022 – Aug. 2022

Tread *Toronto, ON*

- Reduced manual scheduling effort by **50%** by developing a **React-based** calendar management system with **REST API** integrations, automating subcontractor dispatch and availability tracking
- Eliminated payroll disputes by building reusable timesheet dashboard components in **React**, integrated with backend payroll services for accurate time tracking
- Improved post-release stability by identifying and resolving bugs through structured **QA** testing and **Jira-tracked** defect workflows

EDUCATION

Carleton University Ottawa, ON

Honours Bachelor of Computer Science, Stream: Computer & Internet Security *Sept. 2019 – Jun. 2024*

- Awards: Dean's Honour List (2020-2022), University Entrance Scholarship, George Fierheller Scholarship, Gerhard Herzberg Scholarship
- Related Coursework: Object-Oriented Programming, Computing in Python, Data Structures and Algorithms, Database Management Systems, Software Engineering Practices